PSY420M

Advanced Data Analysis and Methods for the Behavioral Sciences

Attendance!

Agenda

In this lab, we'll look at the social approaches vs. hormone data that we saw in the videos.

We will

- Visualize the samples – frequency polygons and/or boxplots
- Visualize the sampling distributions – means and 95% CIs
- Compare oxy vs. estrogen (with a plot and t-test)
- Compare treatment vs. control

Summary of deliverables?

You should submit a lab report including the following:

- A graph of the distributions for each sample (boxplot and/or frequency polygon)
- A graph of means & 95% CIs for each of the 4 groups
- A graph and t-test summary comparing estrogen and oxy
- A graph and t-test summary comparing treatment and control

Rstudio!

- Make a lab 11 folder on your system
- Get the .csv file from canvas called **”hormoneData.csv".**
- Put it into this folder.
- Make a new project called lab 13 and create it in your new lab13 folder

1. First things first, as always, we'll load the tidyverse

```
library(tidyverse)
```

2. Load the data

```
dat <- read_csv("hormoneData.csv")
```

Let’s make the frequency polygon

- `ggplot(dat, aes(x = approaches, color = hormone)) + geom_freqpoly(binwidth = 0.5)`
- The default bins are too narrow! Tweak it to make it look prettier

Let’s make a boxplot too

- Challenge: Can you remember how to do this?
  - Make sure to add jittered points
  - Add some color for extra flair
  - Play around with jittering the points horizontally, and experiment with size, color, and transparency until you get a plot that looks good to you

Make a new variable for treatment vs. control

- ```
  dat <- mutate(dat, treat = case_when(
                     hormone == "V" ~ "Control",
                     hormone == "E" ~ "Treatment",
                     hormone == "O" ~ "Treatment",
                     hormone == "P" ~ "Treatment"))
  ```
  ```

Make a new summary table and plot

```
tVsCSum <- dat %>%
          group_by(treat) %>%
          summarize(
          counts = n(),
          means = mean(approaches),
          sds = sd(approaches))

tVsCSum <- mutate(tVsCSum, ses = sds/sqrt(counts))
tVsCSum <- mutate(tVsCSum, cis = 1.96*ses)
```

Then plot the means and CIs!

Now we make a plot just for E and O

- `Challenge: Can you make a summary table of means containing just E and O hormones?`
  - `(We can start off using filter, as below)`
- `dat %>% filter(hormone == "E" | hormone == "O") %>%`

  `group_by(hormone) %>% ...`

From here try to make a mean & 95% CI plot for E and O

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